



RESEARCH ARTICLE

Effect of phosphorus levels on yield, nutrient uptake and phosphorus utilisation efficiency in rice (*Oryza sativa* L.) using ^{32}P radiotracer technique under pot culture

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Abstract

Phosphorus (P) is a primary limiting macronutrient in tropical and subtropical rice production. While total soil P may be high, over 50 % of global agricultural soils suffer from low bioavailable orthophosphate due to strong chemical fixation with aluminium (Al) and iron (Fe) oxides in acidic soils, or calcium (Ca) in alkaline soils. Developing and deploying P-efficient rice genotypes capable of maintaining growth and yield under low soil P availability is a key priority for sustainable agriculture. A pot culture experiment was carried out at the Radio Isotope Laboratory, Department of Soil Science and Agricultural Chemistry, Tamil Nadu Agricultural University, Coimbatore, Tamil Nadu. The treatment comprised factor A: rice genotypes viz., TNRH-180, CB08504, CB06732 and ADT 47 and factor B: P levels viz., 0, 25 and 50 kg P ha⁻¹. Phosphorus was applied in the form of the radioisotope ^{32}P , the experiment aimed to evaluate and quantify yield performance, phosphorus use efficiency (PUE) and the native P supplying capacity of low P soils. The findings indicated that the rice genotypes viz., TNRH-180 recorded higher yield attributes, yield and PUE viz., productive tillers (8.4 nos.), panicle length (24.1 cm), grains per panicle (167.7 nos.), test weight (17.2 g), grain yield (22.3 g/pot) and PUE (23.72 %) and was followed by CB06732 respectively.

Keywords: nutrient uptake; phosphorus levels; phosphorus use efficiency; ^{32}P radiotracer; rice

Introduction

Rice (*Oryza sativa* L.) is a critical staple food crop for a significant global population. It meets more than 80 % of their dietary needs. However, it typically has a low phosphorus use efficiency (PUE) of around 25 %. Because of this inefficiency, plants exhibit signs of P deficiency, such as stunted growth, fewer branches (tillering) and narrow, short, dark green leaves. Additionally, affected plants may produce fewer panicles and grains and develop thin stems (1). Plants have developed diverse adaptive mechanisms to obtain P from soil, especially in reaction to P scarcity. These changes can be in the form of anatomical, morphological, physiological, biochemical, or genetic changes (2). Some plants can change P that is not soluble into forms that are bioavailable, which makes P more efficient. Under P deficient conditions, the release of nucleases, organic acids, protons and acid phosphatase (APase) makes it easier for P to dissolve, mineralise and desorb, which in turn improves phosphorus acquisition efficiency (PAE) (3). In addition, plants adjust their root system architecture (RSA) such as the root-to-shoot biomass ratio (RSR), lateral root growth and root hair proliferation to improve PAE by extracting P from decomposing organic matter in the soil (4).

Phosphorus is among the seventeen essential nutrients required for plant growth. However, it limits crop productivity on over 40 % of the world's arable land. Global P reserves are being consumed rapidly and some predictions suggest that natural soil P reserves may be exhausted by 2050, posing a serious risk to sustainable agriculture (5). A large proportion of P fertilisers applied to soil often become adsorbed, making them unavailable to plants that lack specific adaptive traits. Even so, the right use of P-based fertilisers can increase the amount of P available in the soil and, as a result, the number of crops that can be grown to meet the needs of the growing world population. Phosphorus may also be recycled through crop residues by putting them back into the soil. When crop residues used as livestock feed, some of the P goes back into the soil through manure or bone meal. Plants have evolved diverse adaptive mechanisms, called P-efficiency strategies, to get enough P when conditions are unfavourable. These mechanisms involve both enhanced uptake efficiency, the plant's capacity to absorb more P under deficient conditions and increased utilisation efficiency, which is the ability to produce a greater biomass yield per unit of absorbed P. Plants enhance P uptake efficiency through multiple mechanisms, including changes in root architecture, expansion of the root system,

formation of longer root hairs and finer roots, secretion of low molecular weight organic acids, protons and enzymes such as phosphatases and phytases, establishment of mycorrhizal associations, development of cluster roots and the activation of high affinity P transporters (6).

Radioisotopes undergo radioactive decay by emitting energy and radiation, which have been extensively used in numerous agricultural applications. The use of radioisotopes has been applied in the development of high-yielding crops to enhance productivity and for studying the dynamics of fertilisers in different plants. In this way, research on agricultural problems has become more efficient and faster. Additionally, the mass production of radioisotopes in nuclear reactors and atomic plants has led to increased applications of radioisotopes in agriculture (7). Since fertilisers comprise an economically important input, efficient fertiliser application plays a very important role in reducing costs associated with crop production. There must be an effort to increase plant nutrient uptake efficiency and reduce nutrient losses to the environment. In this connection, radioisotopes become a valuable means of measuring initial P and nitrogen (N) content in the soil, which is extremely important when determining optimal fertiliser application amounts (8). Therefore, the study aimed to quantify the yield, nutrient uptake and PUE using the radioisotope material in rice crops under the pot culture method.

Materials and Methods

Experimental location and initial study

The first field experiment was conducted in Paddy Breeding Station, Coimbatore with 22 rice genotypes viz., CB06732, TNRH-180, AS06034, CB08536, CB08513, CB08504, CB08509, CB06535, TP08010, AD07038, TM07333, AS06035, AD08010, TM07278, ACM07001,

AS6016, CORH-3, ADT-47, ADT-43, ASD-16, Co-47 and ADT-45 (varieties, pre-release cultures and landraces) at 0 and 50 kg P₂O₅ ha⁻¹ in P deficient soil of Noyyal series. Based on P uptake efficiency, four rice genotypes viz., TNRH-180, CB08504, CB06732 and ADT-47 were selected from the field experiment.

Pot culture experiment

The pot culture experiment was executed at Radio Isotope (Tracer) Laboratory, Department of Soil Science and Agricultural Chemistry, Tamil Nadu Agricultural University, Coimbatore. The experiment soil was filled in earthen pots at 10 kg pot⁻¹, lined with polythene sheets. The treatment consisted factor A: rice genotypes viz., TNRH-18, CB08504, CB06732 and ADT-47 and factor B: fertiliser levels viz., 0, 25 and 50 kg P. This lab experiment was laid out in a completely randomised block design (CRBD) with 3 replications (Fig. 1). Four hills were maintained in pot culture and duration of the experiment was 130 days. Fertilisers were applied as per the ratio of 150 : 50 for N and potassium (K), whereas P was provided in the form of a radioactive isotope of ³²P. Before commencing the experiment, initial soil samples were collected from the field for various physico-chemical properties. After the rice crop was harvested, both grain and straw samples were prepared and subjected to a radio assay. The P from the plant samples, extracted using a triacid mixture, was precipitated as magnesium pyrophosphate (9).

Nature of phosphorus³²

Carrier-free radioactive ³²P was procured from the Board of Radiation and Isotope Technology (BRIT), Mumbai, in the form of orthophosphoric acid dissolved in dilute hydrochloric acid and used for the experiments. The isotope has a physical half life of 14.3 days and undergoes β-emission (negatron release) with a maximum energy of 1.71 MeV, transforming into sulphur.



Fig. 1. General view of pot culture experiment.

Preparation of pots and tagging of phosphorus³²

Soil samples with low available P content were collected from the Paddy Breeding Station and used for a pot culture experiment. Each pot was filled with 10 kg of wet soil. All pots received a uniform supply of N (as urea) and K (as muriate of potash), which were applied in four equal splits at the basal, active tillering, panicle initiation and flowering stages. Phosphorus³² was supplied as labelled single superphosphate (SSP), applied basally. The labelled SSP was prepared by thoroughly mixing rock phosphate, the ³²P stock solution and concentrated sulfuric acid (H₂SO₄) in calculated proportions with constant stirring.

The prepared material was dried under an infrared lamp and then finely ground using a mortar and pestle. The radioactive fertilisers were applied to the pots following all prescribed safety measures for handling isotopes. After application, the pots were irrigated and maintained in a submerged condition and seedlings raised in the nursery for 22 days were transplanted into each pot at of 2–3 seedlings per pot.

Radioisotope handling and safety protocol

Radioisotope handling must follow strict radiation safety principles to minimise radiation exposure and prevent contamination. Since ionising radiation can cause biological harm, exposure should be maintained as low as reasonably achievable (ALARA) by minimising the quantity of radioactive material handled, reducing exposure time, maximising the distance from the radiation source and using appropriate shielding. Radioisotopes should be classified according to their toxicity based on the type and energy of radiation, physical and biological half-life, absorption pattern and the quantity and chemical form handled. Radiation safety practices should comply with the recommendations of the international commission on radiological protection (ICRP), including justification of use, optimisation of protection and adherence to prescribed dose limits for occupational workers and the public. Personnel handling radioisotopes should wear appropriate personal protective equipment, including laboratory coats, disposable gloves, closed footwear or shoe covers and other protective clothing as required. Personal radiation exposure should be monitored using thermoluminescent dosimeter (TLD) badges or film badges, which are typically evaluated at regular intervals (commonly once every three months) by authorised radiation safety personnel to ensure that occupational dose limits are not exceeded.

Radiation exposure and contamination should be continuously monitored using pocket ionisation chambers, survey meters, Geiger counters, scintillation counters and contamination monitors. Airborne contamination should be assessed using static air samplers, while surface contamination should be monitored either by direct measurement with portable contamination monitors or by indirect smear tests using filter paper or cotton swabs. In the event of contamination, immediate decontamination measures should be undertaken by removing contaminated clothing, gently washing the affected skin with soap and water and cleaning laboratory equipment and work surfaces with water followed by suitable detergents or approved decontaminating agents such as ethylenediaminetetraacetic acid (EDTA), chromic acid, or dilute mineral acids. Radioactive spills should be contained using absorbent materials and all contaminated materials should be treated as radioactive waste until contamination levels are reduced to permissible limits. All radiation work should be performed only by

trained personnel following approved radiation safety protocols and ALARA principles to ensure the protection of personnel, the laboratory environment and the surrounding ecosystem.

Waste disposal of radioisotope materials

The disposal of radioisotope materials should be carried out in accordance with the Atomic Energy (Safe Disposal of Radioactive Wastes) Rules, 1987, with the primary objective of ensuring that radiation exposure to personnel and the environment remains within prescribed limits. Radioactive waste generated during research activities should be segregated at the point of generation based on its physical form, activity level, half-life and chemical characteristics. The guiding principles of radioactive waste management include concentrate and contain, delay and discharge and dilute and discharge, with the disposal method selected according to the characteristics and hazard associated with each radionuclide. Short-lived solid radioactive wastes should be collected in designated containers and disposed of through controlled burial in approved pits, whereas small quantities of short-lived liquid wastes that are soluble or dispersible in water may be discharged into the sanitary sewage system within the prescribed regulatory limits. Liquid scintillation cocktails and other organic radioactive waste should be managed as chemical waste rather than discharged into sewage systems. Radioactive animal carcasses and contaminated plant materials should be securely wrapped in polythene bags and disposed of using the same controlled burial procedures as solid radioactive waste. Incineration of combustible radioactive waste may be undertaken only in suitable facilities under controlled conditions, with the resulting ash collected and disposed of as solid radioactive waste. Long-lived and non-dispersible radioactive wastes should not be disposed of locally and must be managed under the guidance and authorisation of the Radiological Safety Division of the Atomic Energy Regulatory Board (AERB). Throughout the waste management process, proper segregation, documentation, monitoring and compliance with regulatory limits are essential to ensure safe and environmentally responsible disposal of radioactive materials.

Short-lived solid radioactive waste must be separated from normal trash right at the worktable and buried locally in custom-made dirt pits. Pits must be exactly 120 × 120 cm. When filled, the waste must be covered by a top layer of compact soil that is at least 120 cm thick. If multiple pits are needed, they must be spaced at least 180 cm apart. A facility is allowed to dig a maximum of 12 pits per year. The area must be locked, fenced off and labelled with warning signs. Each pit has a strict radioactive limit depending on the material. For example, a single pit can hold up to 9250 MBq of tritium ³(H), 370 MBq of ³²P and 37 MBq of iodine-13. After waiting for 7 to 10 half-lives, the radioactivity decreases enough for the waste to be excavated and safely disposed of in a normal municipal dump. This allows the pit to be reused.

Plant sample analysis

Plant samples were collected at the time of harvest and dried in a hot air oven at 65°C, after which their dry weight was recorded. The oven-dried materials were chopped, finely ground using a Willey mill and stored in wide-mouthed, stoppered bottles. Representative subsamples were then taken and analysed for N, P, K, calcium (Ca), magnesium (Mg) and micronutrients following standard analytical methods. Phosphorus present in the plant's triacid extract was determined by precipitating it as magnesium pyrophosphate,

McKenzie and Dean used as end-window Geiger Muller counter (Type para nuclear counter1a, Hyderabad). The counting efficiency of the counter for ^{32}P was 7%.

Interpretation of radio assay data

Radio assay data were converted into different parameters by using following formula:

Absolute activity (disintegrations per second)

The corrected count rate (counts per second, cps) is converted to Becquerels (Bq) or disintegrations per second (dps) by adjusting detector efficiency:

$$\text{Activity (Bq or dps)} = \text{Corrected count rate (cps)} / \text{Detector efficiency (\%)} \times 100$$

Specific activity

Specific activity represents the radioactive decay rate per unit mass of total P present in the digested tissue:

$$\text{Specific activity (dps mg}^{-1}\text{P)} = \text{Sample disintegration rate (dps)} / \text{Total P mass in sample (mg)}$$

Phosphorus derived from fertiliser (% Pdff)

The proportion of total plant P translocated directly from the applied ^{32}P labelled fertiliser standard is calculated as follows:

$$\text{Phosphorus derived from fertiliser (\%)} = \text{Specific activity of plant sample} / \text{Specific activity of fertiliser standard} \times 100$$

Phosphorus derived from soil (% Pdffs)

The fraction of plant P sourced from native soil pools is determined by subtracting the fertiliser-derived fraction from the total P content.

$$\text{Phosphorus derived from soil (\%)} = 100 - \% \text{ Pdff}$$

Where, Pdff is phosphorus derived from fertiliser.

Fertiliser phosphorus uptake

The absolute mass of P accumulated by the plant from the exogenous fertiliser source per experimental unit (pot) is expressed as follows:

$$\text{Fertiliser P uptake (mg P pot}^{-1}\text{)} = \% \text{ Pdff} / 100 \times \text{Total P uptake (mg P pot}^{-1}\text{)}$$

Where, Pdff is phosphorus derived from fertiliser.

Fertiliser phosphorus utilisation efficiency (%)

The percentage of applied fertiliser P successfully recovered by the plant tissue is calculated using the following ratio:

$$\% \text{ P utilisation} = \text{Fertiliser P uptake (mg P pot}^{-1}\text{)} / \text{Total P applied via fertiliser (mg P pot}^{-1}\text{)} \times 100$$

Statistical analysis

The observations recorded for different traits in the study were subjected to statistical analysis (10). The experimental data were further analysed using the procedure described earlier (11). Whenever significant differences among treatments were detected, critical differences (CD) were calculated at the 5 % probability level and reported accordingly. Genotypic clustering in the field experiment was performed using MINITAB software (version 22.5) (12).

Results and Discussion

Yield attributes

Productive tillers per plant in rice genotypes revealed significant

differences between the genotypes. Among the rice genotypes, productive tillers per plant showed gradual increase when applied with higher level of P application (6.1–8.4). The mean value of productive tillers per plant was higher in TNRH-180 (7.9 nos.) followed by ADT-47 (7.9) and the lowest value recorded in CB06732 (6.6). Even at omission P plots, TNRH-180 and ADT-47 performed better under P stress soil (Fig. 2). This may be attributed to variations in enzyme and hormone activity, leading to improved production of productive tillers (8). Rice genotypes can be able to use both available and bound forms of native P by releasing organic acids from their roots, which help dissolve the nutrient. Consequently, the identified genotypes have the capacity to improve PUE in soil ecosystems under P stress (13).

Low P soil significantly increased the number of productive tillers, root biomass, root-shoot ratio, root bleeding rate and root acid phosphatase (RAP) activity. Total root length, root oxidation activity (ROA) and root total and active absorbing surface areas were significantly increased in strong-tolerance varieties under low P, whereas the opposite responses were observed in weak tolerant varieties. Total root length, ROA, root bleeding rate, root active absorbing surface area and RAP activity were positively and significantly correlated with grain yield, P translocation efficiency and internal P efficiency. These results suggest that the tolerance of rice varieties to a low-P growth condition is closely associated with root growth with higher biomass and activity (14).

The study was to screen 25 rice cultivars at low, medium and high levels of soil P. Number of tillers, root length, plant height, root dry weight and shoot dry weight were related to tissue P concentrations, P uptake and P-use efficiency. Shoot weight was found to be the plant parameter most sensitive to P deficiency. Significant cultivar differences in P use efficiency were found. Phosphorus use efficiency was higher in roots than shoots and decreased with increasing levels of soil P. Positive correlations were found among growth parameters such as plant height, tillers, root and shoot weight and P content of roots and shoots. These results indicate selection of rice cultivars for satisfactory performance under low P availability can be carried out using shoot and root dry weight as criteria (15).

Among the genotypes, TNRH-180 recorded the highest mean panicle length (24.1 cm), while ADT-47 registered the lowest (21.8 cm). The interaction effect indicated that TNRH-180 produced the maximum panicle length (24.5 cm), followed by 24.1 cm when supplied with 25 kg P. This effect may be due to the development of longer panicles bearing a greater number of grains per panicle under higher P application, supported by increased uptake of major nutrients during the reproductive stage (16).

The higher mean number of grains per panicle recorded in rice genotype TNRH-180 (159.1 nos.) and it was followed by CB08504. Among the various P levels, a higher number of grains per panicle was registered in TNRH-180 at 50 kg P. The interaction between genotypes and P levels was significant. This may be attributed to spikelet formation into well-filled grains because of the application of P fertiliser in conjunction with organic manure, enabling the crops to stay green even during the late stages of their life cycle. This would enable photosynthetic assimilation to be more effective, resulting in an increased number of filled grains per panicle (17). Adequate supply of micronutrients was essential for efficient utilisation of major plant nutrients that would react on yield components and

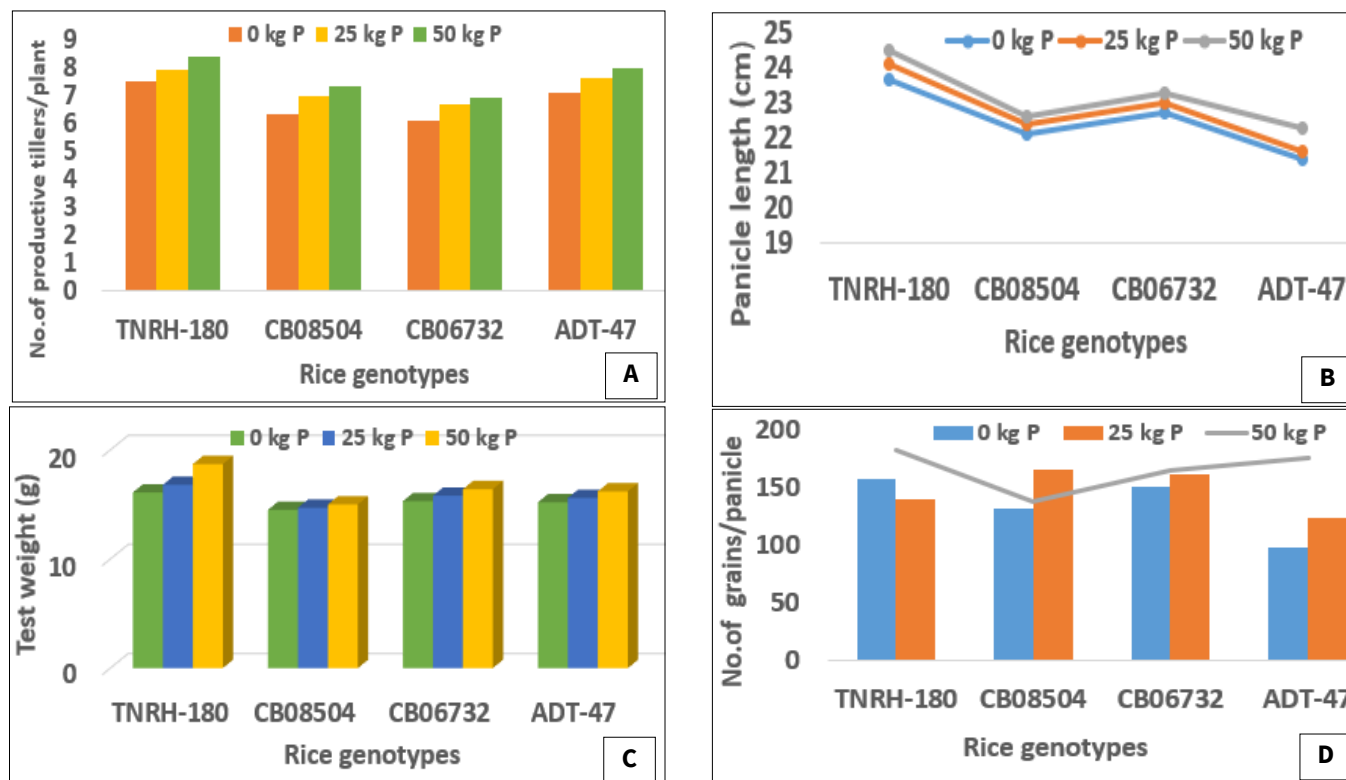


Fig. 2. Yield attributes of the rice genotypes as influenced by different P levels under P stress conditions. (A) number of productive tillers/plant; (B) panicle length; (C) test weight; (D) number of grains/panicle. P: phosphorus.

yield. Better assimilation of photosynthates due to increased application of P and ferrous sulfate (FeSO_4) and better utilisation of incident solar radiation might have increased the size of the sink and effective translocation of the assimilates which in turn led to improved productive tillers per metre, panicle per metre, total grains per panicle, filled grains per panicle and test weight (18).

Thousand grain weights in rice genotypes revealed distinct differences between the genotypes. Among the rice genotypes, thousand grain weight showed gradual increase with enhanced level of P application (15.0–18.7 g). The mean value of thousand grain weight was higher in TNRH-180 (17.2 g) followed by CB06732 (15.9 g) and the lowest value was recorded in CB08504 (14.7 g). This could be attributed to the higher accumulation capacity of the rice genotypes under an adequate supply of macro and micronutrients, which ultimately enhanced the thousand grain weight (19).

Test weight, being genetically controlled and it can be influenced only to a limited effect by agronomical means. It was influenced by P and iron (Fe) application. Higher accumulation ability of the hybrid rice with optimum supply of macro and micronutrients increased the test weight (20). High level of P application resulted in higher grain weight which could be due to higher nutrient uptake and partitioning of photosynthates for grain development (21).

Grain and straw yield

Among the rice genotypes, grain yield showed a gradual increase when applied with higher level of P (19.1 to 25.0 g pot^{-1}). The mean value of grain yield was higher in TNRH-180 (23.1 g pot^{-1}) followed by CB06732 (22.3 g pot^{-1}) and the lowest value was recorded in CB08504 (18.7 g pot^{-1}). Comparatively, when P was applied at different doses viz, 0, 25 and 50 kg P the rice genotypes like TNRH-180, CB06732 and CB08504 found superior in grain yield under different P conditions (Fig. 3). It might be due to the enhancement of yield parameters like

panicles per metre, panicle length, panicle weight, grains per panicle and filled grain percentage, consequently increasing grain yield (22). Even the plot received 0 kg P TNRH-180 recorded higher grain yield (21.0 g pot^{-1}) followed by CB06732 respectively. It may be due to rice genotypes that can use both available and bound forms of native P by releasing organic acids from their roots, which help dissolve the nutrient. Consequently, the identified genotypes have the capacity to improve P utilisation efficiency under P soil ecosystem. Low P tolerant rice genotypes (TNRH-180) possess "metabolic flexibility" and adapt to P omission by altering root physiological traits and exudation patterns to mobilise bound rhizosphere P.

Straw yield increased with higher P application, ranging from 24.0–32.3 g pot^{-1} . The highest mean straw yield was observed in CB06732 (30.1 g pot^{-1}), while the lowest was recorded in CB08504 (26.5g pot^{-1}). The interaction between genotypes and P levels was significant, with CB06732 producing the maximum straw yield (34.0g pot^{-1}), followed by ADT-47 (33.1g pot^{-1}), whereas the lowest yield (22.9 g pot^{-1}) was noted in CB08504 under P omission conditions. The rise in grain and straw yield could be attributed to the overall enhancement of plant growth, as P plays a crucial role in plant metabolism, ultimately leading to improved yield (23).

Plant nutrient uptake

Nitrogen uptake

Among the rice genotypes, N uptake showed gradual increase with higher level of P application and the range varied from 247–523 mg pot^{-1} . The mean value of N uptake was higher in TNRH-180 (433 mg pot^{-1}) followed by CB06732 (408 mg pot^{-1}) and the lowest value was recorded in CB08504 (347.3 mg pot^{-1}) (Fig. 4). This may be explained in terms of the effect that organic manures have on the loss of N and formation of organo-mineral complexes, ensuring a constant supply of N for the rice crops. The higher uptake of major nutrients in plants may be attributed to greater dry matter

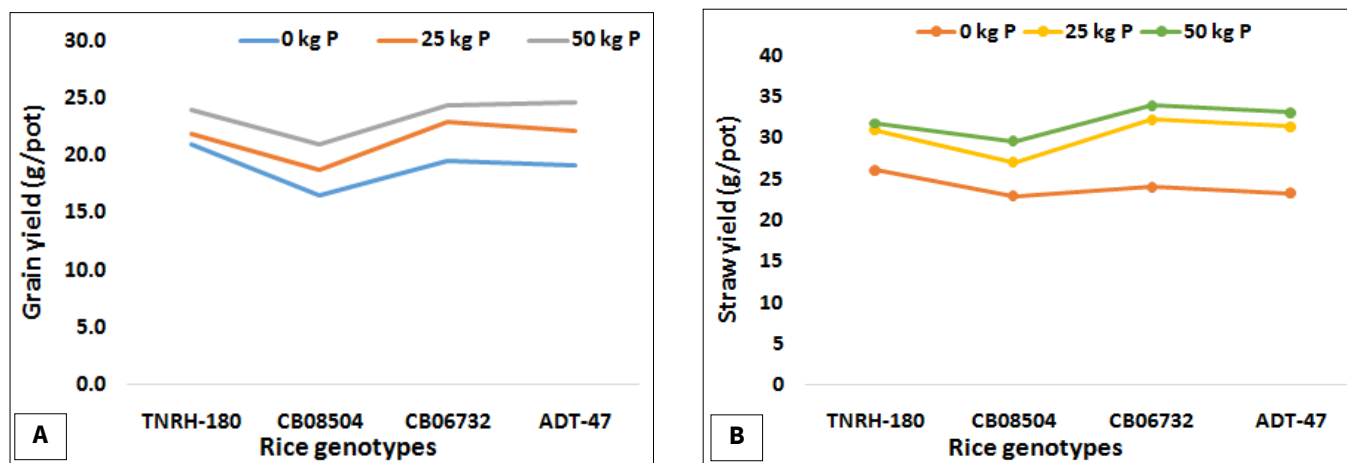


Fig. 3. Yield (g/pot) of the rice genotypes as influenced by different P levels under P stress condition. (A) grain yield; (B) straw yield.

P: phosphorus.

accumulation along with an increased concentration of N in different plant parts. In contrast, the higher N uptake in transplanted rice may be due to greater grain and straw yields, which resulted in higher N removal from the soil. This might also be associated with better root development that enhances nutrient availability and absorption from the soil (24). Nitrogen uptake was increased with application of 80 kg P combined with 50 kg FeSO_4 . It may be due to the ability of hybrid and application of adequate amount of N greatly induces the uptake. Increasing levels of N, P, K application influenced the total N, P and K uptake at harvest of rice (25).

Phosphorus uptake

Phosphorus uptake increased significantly and progressively with higher levels of P application, ranging 127–226 mg pot^{-1} (Fig. 3). Even under P-omitted plots, the rice genotypes like CB06732, TNRH-180 registered higher P uptake in low P soil. This could be due to an increase in P availability in the soil resulting from fertilisation and the subsequent rise in P levels in the rice plants. The increased P content also contributed to increased dry matter accumulation (26). The higher P uptake in transplanted rice might be attributed to higher yield and better root growth, which improved soil aeration and enhances nutrient availability and absorption from the soil (27).

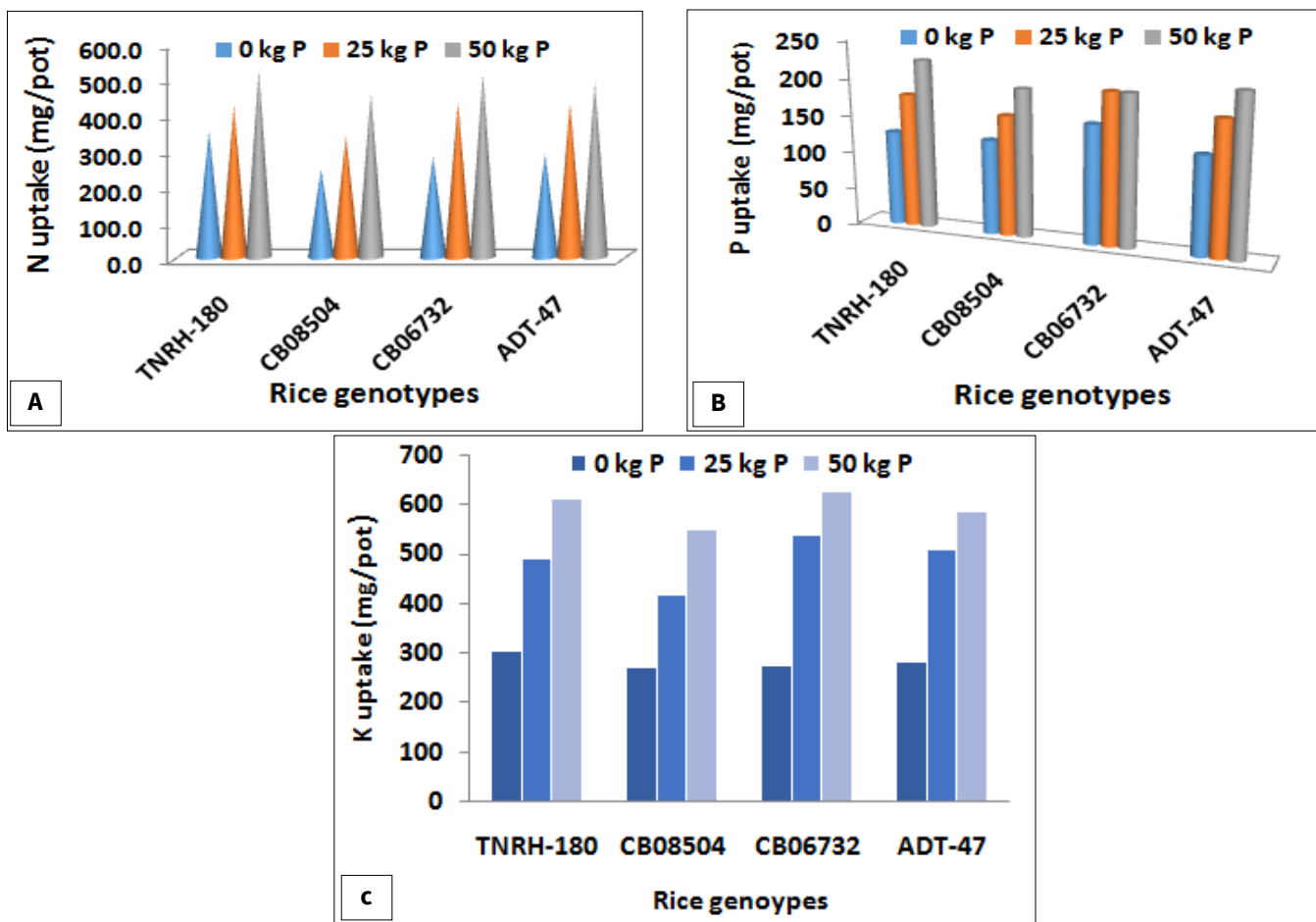


Fig. 4. Total plant nutrient uptake (mg/pot) of the rice genotypes as influenced by different P levels under P stress condition. (A) N uptake; (B) P uptake; (C) K uptake.

P: phosphorus; N: nitrogen; K: potassium.

A nutrient pool of P and Fe formed on the roots to forage for more inorganic phosphorus (Pi) under low-Pi conditions. Severe long-term low-Pi stress promoted primary root elongation and inhibited lateral root growth, which increased the longitudinal absorption zone of the roots. The decrease in P content of the roots upregulated the expression of the APase gene and increased APase activity. The high-affinity phosphate transporter (Pht1) genes were also regulated significantly. These morphological and physiological responses significantly increased Pi uptake rate and P utilisation efficiency at the melon seedling stage (28).

Potassium uptake

Potassium uptake in rice genotypes showed significant variation across different genotypes and P levels. The K uptake showed a gradual increase with higher level of P application and the range varied from 270–628 mg pot⁻¹. The mean value of K uptake was higher in TNRH-180 (469 mg pot⁻¹) followed by CB06732 460 mg pot⁻¹ and the lowest value was recorded in CB08504 (412 mg pot⁻¹) (Fig. 3). The increased uptake of K in rice genotypes may be due to efficient nutrient uptake by combining the application of organic fertiliser together with N, P and K (29). The application of increased nutrient levels led to an increase in nutrient availability, leading to improved yields and quality (30). Phosphorus and Fe nutrition had a significant influence on K uptake by the crop. Application of 80 kg P with 50 kg FeSO₄ ha⁻¹ registered higher K uptake at all the stages. The higher K recovery might be due to higher drymatter production (DMP) particularly in straw, bulk of the absorbed K is retained (31). The K uptake by rice increased with advancement of crop growth. The highest uptake of K was observed when 100 % P was applied. This might be due to the increased root activity resulting in the increased dry matter accumulation and high utilisation of K (32).

Correlation matrix

The significant relationships among nutrient uptakes (N, P and K) implies that proper fertilisation and nutrient balance in the soil is necessary for optimal crop production. The significant positive association between grain and straw yields and nutrient uptake indicates the need for nutrients such as N and P to determine economic and biomass yields. The moderate relationship between productive tillers and nutrient uptake indicates that nutrients play a key role in vegetative growth and branching, which could affect the yield potential. On the contrary, the insignificant relationships between panicle length and other variables imply that it might not be an important factor in yield determination (Fig. 5).

The moderate correlation between test weight and nutrient uptakes suggests that the filling and quality of the grain are affected by the nutrient availability but not as much as the total yield. It is worth noting that the low correlation between the panicle length and number of grains per panicle and the grain yield shows that the grain yield is more dependent on the number of productive tillers and nutrient availability than on panicle properties. Nutrient management and integration are important to optimise the grain yield.

Percent phosphorus utilisation

There was variation in the mean PUE in the different genotypes of rice, with the highest percentage of P utilisation being observed in TNRH-180 (24 %) when P was applied at two concentrations followed by CB06732. When P was applied at the higher concentration (50 kg P), the genotype ADT-47 (21 %) showed a higher percentage of PUE than the others (Fig. 6). At the lower concentration (25 kg P), the genotype TNRH-180 (29 %) had the highest PUE compared to the others. The decrease in percent phosphorus utilisation (PPU) at 50 kg ha⁻¹ can be attributed to the increased mineralisation of P from the soil (33).

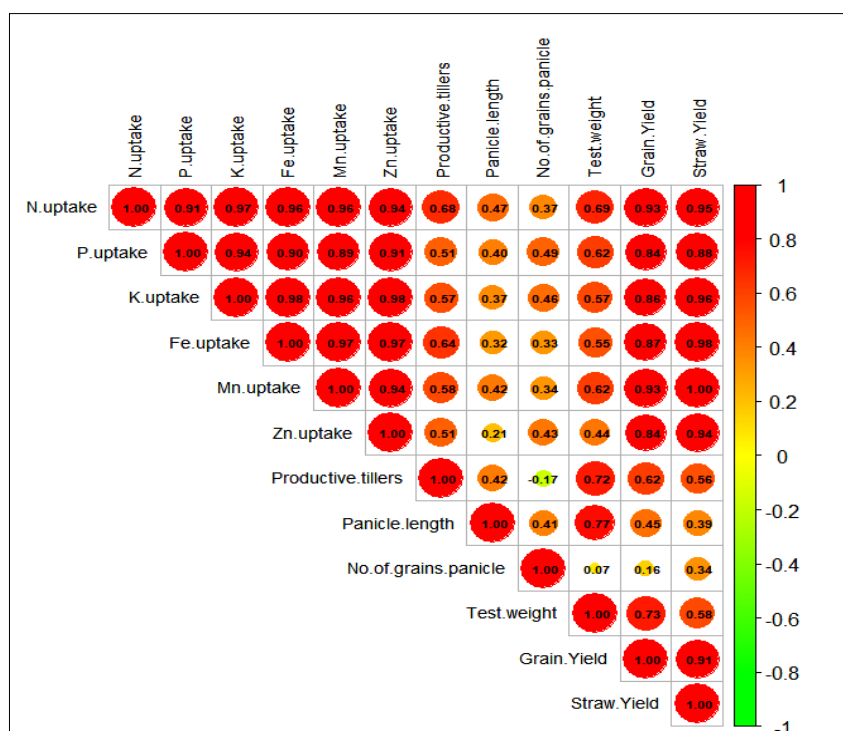


Fig. 5. Graphical representation of correlation matrix (using the Spearman correlation coefficient) among nutrient uptake, agronomic traits and yield parameters.

Red indicates positive correlations, green indicates negative correlations and the intensity and size reflect the strength of correlations. Empty cells represent non-significant correlations ($p > 0.05$). N: nitrogen; P: phosphorus; K: potassium; Fe: iron; Mn: manganese; Zn: zinc.

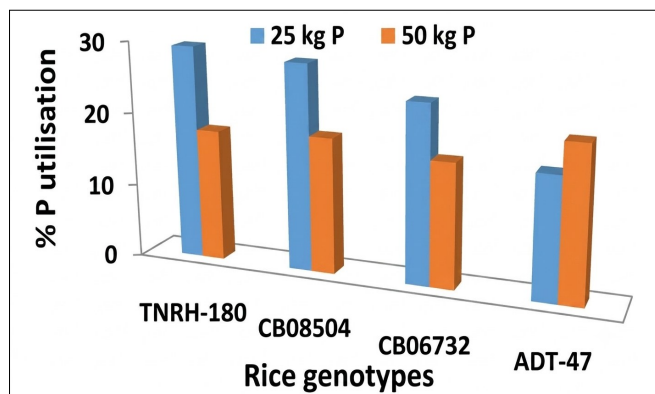


Fig. 6. Percent phosphorus utilisation in post harvest soil sample of the rice genotypes as influenced by different P level under P stress condition.

P: phosphorus.

Conclusion

This study demonstrates that phosphorus (P) application significantly enhances the yield attributes, biomass production and macronutrient (nitrogen (N), P and potassium (K)) uptake of rice genotypes. Among the evaluated cultivars, TNRH-180 emerged as the most superior genotype, consistently recording the highest grain yield, nutrient uptake and phosphorus utilisation efficiency (PUE) achieving a peak percent phosphorus utilisation (PPU) of 29 % at the 25 kg P ha⁻¹ level. Notably, both TNRH-180 and ADT-47 exhibited robust performance and tolerance under P-deficient (omission) conditions, which is closely linked to adaptive root morphology and enhanced physiological activity. These genotypes proved a strong capacity to efficiently utilise both native soil P and applied P from soil. Therefore, they are recommended for cultivation in low P regions of soil ecosystem. Thus, utilising such scientific approaches will enable farmers to cultivate rice genotypes suitable for P stress, which will reduce P fertiliser usage and prevent soil pollution. Future research, cultivar development can be accelerated by using metabolomics to isolate the specific organic acids responsible for unlocking bound native P, alongside molecular screening to map the underlying high affinity phosphate transporter genes.

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Authors' contributions

VS, RR and TB contributed to the research activities, field establishment and drafting of the research article. MV, MM and BBS reviewed and proofread the manuscript, performed statistical analysis on the data collected during the research, assisted with the analysis process and participated in sequence alignment. All authors read and approved the final manuscript.

Compliance with ethical standards

Conflict of interest: Authors do not have any conflict of interests to declare.

Ethical issues: None

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